

EnsDiffAD: Ensemble Diffusion Models for Multivariate Time Series Anomaly Detection

Qian Ma , Yanyang Li, Mei Bai , Xite Wang , Shikai Guo , Yu Gu , and Ge Yu , *Senior Member, IEEE*

Abstract—Multivariate time series anomaly detection (MTSAD) is critically important across a wide range of research fields and applications. However, achieving accurate anomaly detection in multivariate time series (MTS) presents several challenges. First, existing MTSAD methods mainly emphasize modeling long short-term temporal dependencies, while neglecting the intricate and strongly correlated inter-variable dependencies, which limits detection accuracy. Second, distributional shifts caused by external factors further hinder a single model's ability to generalize across evolving patterns, risking missed subtle anomalies and compromising detection robustness. To address these issues, we propose EnsDiffAD, a novel MTSAD framework leveraging an ensemble of improved diffusion models. A key component of our framework is TiT-Diffusion, a denoising architecture designed to model the entangled yet distinct nature of temporal and inter-variable dependencies. Instead of directly combining existing models, TiT-Diffusion decouples these aspects into complementary modules: a temporal Transformer captures long-term and short-term temporal patterns, while an inter-variable iTransformer models global cross-variable correlations. This decomposition mitigates representation conflicts and facilitates more expressive and disentangled feature learning. Additionally, we enhance dynamics and uncertainty modeling via a novel loss function and ensemble multiple TiT-Diffusion models during training. Experiments on five real-world datasets demonstrate EnsDiffAD's superiority.

Index Terms—Anomaly detection, Multivariate time series, Deep learning, Ensemble learning, Generative Networks.

I. INTRODUCTION

MULTIVARIATE time series anomaly detection (MTSAD) is critically important across a wide range of research fields and applications, including finance, manufacturing and energy management.

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Qian Ma, Yanyang Li, Mei Bai, Xite Wang, and Shikai Guo are with the College of Information Science and Technology, Dalian Maritime University, Dalian 116026, China (e-mail: maqian@dlmu.edu.cn; liyanyang@dlmu.edu.cn; baimei@dlmu.edu.cn; wangxite@dlmu.edu.cn; shikai.guo@dlmu.edu.cn).

Yu Gu and Ge Yu are with the Department of Computer Science, Northeastern University, Shenyang 110819, China (e-mail: guyu@mail.neu.edu.cn; yuge@mail.neu.edu.cn).

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Fig. 1. Example cases of ImDiffusion and EnsDiffAD Models for Anomaly Detection on the MSL Dataset with its true labels (blue). Predicted labels (red) for this dataset.

The presence of anomalies in multivariate time series (MTS) may indicate the occurrence of novel or unforeseen events, such as production failures [1], [2], system malfunctions [3], [4], or medical conditions like arrhythmias [5]. Consequently, accurate anomaly detection is essential for facilitating further investigation. However, achieving accurate anomaly detection in multivariate time series (MTS) data presents several challenges.

(1) A fundamental challenge in MTSAD lies in effectively modeling both temporal dependencies and inter-variable dependencies. Existing approaches have explored different paradigms: Graph-based models [6], [7] employ graph structures to facilitate information propagation across variables, whereas Transformer-based models [8] specialize in capturing long-range temporal dependencies. More recently, several diffusion-based models, such as ImDiffusion [9] and DiffAD [10], have been proposed for the MTSAD task, achieving state-of-the-art detection results by exploiting the strong ability of diffusion models [11] to model complex joint distributions through iterative denoising. However, these methods primarily emphasize modeling temporal dependencies, while falling short in effectively capturing the complex inter-variable dependencies, or conflate temporal and inter-variable dependencies within a single architecture. As a result, this oversight may lead to suboptimal detection accuracy when anomalies are primarily caused by complex inter-variable interactions rather than by dominant temporal patterns. As illustrated in Fig. 1, ImDiffusion fails to clearly distinguish between anomalous and normal intervals in its reconstruction residuals due to its limited capability in capturing complex inter-variable dependencies. In contrast, EnsDiffAD, which models inter-variable dependencies via the iTransformer

TABLE I
NOTATION TABLE

Symbol	Description
t	Diffusion step index
T	Total number of diffusion steps
$\text{PE}(t)$	Sinusoidal positional encoding of diffusion step t
$\mathbf{e}_t$	Diffusion step embedding
ω_i	Frequency parameter of the i -th sinusoidal component
w^i	Modulating weight for the i -th TiT-Diffusion model
γ	sharpening coefficient
Y_t	Step-level anomaly indicator vector at diffusion step t
y_l	Final anomaly label at time point l

module, produces higher residuals during anomalies, enabling more accurate anomaly identification.

(2) Multivariate time series generated by modern systems introduce additional challenges due to their susceptibility to external influences. Specifically, the MTS often exhibits instability, with data distributions potentially shifting over time, necessitating a high level of adaptability from models. A single model may struggle to perform well amidst these distributional shifts, as its generalization ability is limited by both its complexity and the diversity of the data [12]. This limitation may result in the model overlooking subtle or localized patterns, especially in the presence of weak or transient anomalies. Consequently, such constraints not only impair the model's generalization capabilities but also increase the risk of instability or overfitting, posing significant challenges to achieving robust and reliable anomaly detection results. While ensemble learning is a natural strategy for enhancing robustness, most existing anomaly detection frameworks either apply ensemble mechanisms post-hoc or fail to tailor them to diffusion-based generative models, neglecting collaborative error correction among ensemble members during training.

To address the challenges above, we propose EnsDiffAD, a novel MTSAD framework that integrates multiple strategies to enhance model capability. Specifically, we first extend the traditional diffusion model into TiT-Diffusion by incorporating Transformer [13] and iTransformer [14] networks into the denoising process. TiT-Diffusion leverages the Transformer to capture temporal dependencies and the iTransformer to model fine-grained inter-variable correlations. Second, EnsDiffAD employs ensemble learning to integrate multiple TiT-Diffusion models. A novel ensemble loss enables each model to learn from the overall distribution while correcting errors from its predecessors, promoting inter-model knowledge sharing. By integrating TiT-Diffusion and ensemble learning, EnsDiffAD robustly captures the complex and dynamic nature of MTS data, improving predictive performance, robustness, and generalization.

In summary, our major contributions are as follows:

- We propose EnsDiffAD, a novel anomaly detection framework that ensembles multiple enhanced TiT-Diffusion models, specifically designed for multivariate time series.
- We propose TiT-Diffusion, a dual-architecture diffusion model that explicitly decouples temporal and inter-variable dependencies, enabling more expressive and disentangled

feature learning for multivariate time series anomaly detection.

- We propose an ensemble strategy for integrating multiple TiT-Diffusion models, coupled with a novel loss design, to enhance anomaly detection accuracy and improve overall generalization in multivariate time series.
- We compare the proposed EnsDiffAD with 15 state-of-the-art anomaly detection baselines over five real-world benchmark datasets. The results indicate that EnsDiffAD achieves state-of-the-art performance in both anomaly detection accuracy and timeliness.

II. PRELIMINARIES

A. Problem Statement

1) *Multivariate Time Series*: We consider a multivariate time series (MTS) as a sequence of observations arranged in chronological order, where each observation is a multi-dimensional vector containing multiple variables (features). At each time point, the observation is represented as $X_l = (x_{l,1}, x_{l,2}, \dots, x_{l,K}) \in \mathbb{R}^K$. Thus, a MTS is defined as $X = (X_1, X_2, \dots, X_L) \in \mathbb{R}^{L \times K}$, where K is the number of variables, and L is the number of time steps.

We also denote a mask matrix $M \in \mathbb{R}^{L \times K}$ where each element $m_{l,k} \in \{0, 1\}$ indicates whether the corresponding data point $x_{l,k}$ is visible (unmasked) or missing (masked). In the context of a multivariate time series X , applying the mask matrix M allows us to generate a partially masked multivariate time series $\tilde{X}$, formally denoted as

$$\tilde{X} = X \odot M = \begin{cases} x_{l,k} & \text{if } m_{l,k} = 1 \\ 0 & \text{if } m_{l,k} = 0, \end{cases} \quad (1)$$

where $\odot$ denotes the Hadamard product. The partially masked time series $\tilde{X}$ is employed during model training to simulate missing values or partially obscure specific observations. This approach aims to enhance the model's robustness against noise and data incompleteness.

2) *MTSAD Task*: Given a multivariate time series X and its corresponding ground-truth anomaly label sequence Y , our objective is to learn a mapping $A(X) \rightarrow \hat{Y}$, where $\hat{Y} \in \mathbb{R}^L$ represents the predicted anomaly labels for each time step. Here, $\hat{y}_l \in \{1, 0\}$ indicates whether the observation $X_l \in X$ at time l is anomalous (1) or normal (0). An effective anomaly detection model should accurately distinguish anomalous patterns from normal behaviors, even under noise, temporal dependencies, and high dimensionality.

For clarity, Table I summarizes the key notations used across different modules of EnsDiffAD.

B. Related Work

1) *Time Series Anomaly Detection*: Traditional MTSAD methods are generally categorized into four types: 1) Statistical-based methods model data distributions to identify anomalies. 2) Clustering-based methods detect anomalies via distances to cluster centroids [15]. 3) Distance-based methods rely on pairwise distances among samples. 4) Density-based methods identify

anomalies by estimating local data density [16]. However, these approaches often struggle with high-dimensional and complex multivariate time series.

With the rise of deep learning, neural networks have been widely adopted to capture spatio-temporal dependencies [17]. Existing methods can be broadly divided into: 1) Prediction-based methods, which detect anomalies by comparing predicted and observed values. Models such as RNNs [18], LSTM [19], and Transformers [20], [21], [22] are commonly used. 2) Reconstruction-based methods, which identify anomalies via reconstruction errors (e.g., CATCH [23], H-PAD [24]). 3) Representation learning-based methods, which learn informative features for detection, including GNN-based [6], [25] and Transformer-based models like TS2Vec [26]. 4) Hybrid methods, which combine prediction, reconstruction, and representation learning [7], [27], [28], such as FuSAGNet [27].

2) *Ensemble Learning*: Ensemble learning is a well-established strategy for improving robustness and generalization by combining multiple models with complementary behaviors [29], [30], and has been shown to be effective in handling evolving dynamics and structural changes commonly observed in real-world time series [12], [31]. In time series analysis tasks, traditional ensemble strategies, such as output-level averaging, train multiple anomaly detection models independently and aggregate their outputs via simple averaging or voting schemes [12], [32], [33]. For example, pTSE [33] ensembles multiple probabilistic forecasting models by aggregating their output distributions in a post-hoc manner, demonstrating improved robustness under non-stationary dynamics. Beyond static averaging, adaptive ensemble strategies have been explored to dynamically adjust the contribution of base models based on local performance, temporal regimes, or historical errors [12], [34].

Despite recent advances, most MTSAD methods, such as DCdetector [35] and TranAD [8], primarily focus on temporal dependencies while overlooking complex inter-variable correlations. In addition, handling non-stationary dynamics remains challenging. Ensemble learning has been explored to improve robustness and generalization; however, existing strategies mainly rely on post-hoc aggregation and lack training-time collaboration or progressive error correction among ensemble members. This limitation is particularly critical for diffusion models, where errors may accumulate across denoising steps. To address these issues, we propose an ensemble diffusion framework that integrates Transformer and iTransformer architectures to jointly model temporal and inter-variable dependencies. Furthermore, a loss-level ensemble mechanism is introduced to enable progressive, error-aware collaboration among multiple diffusion models, leading to more adaptive and robust anomaly detection.

III. ENSDIFFAD FRAMEWORK

A. Overview of EnsDiffAD

As shown in Fig. 2(a), the proposed EnsDiffAD mainly consists of four components:

1) *Data Masker*: Pre-processes the input X by randomly masking certain observed values to generate a binary mask M and the partially masked input $\tilde{X}$. This serves two purposes: (1) it encourages the model to better learn the underlying data distribution from unmasked values; (2) it mitigates the impact of unmasked anomalies during inference, which may blur reconstruction errors between normal and abnormal points. To address this, we adopt imputation errors (i.e., reconstruction errors on masked data) for anomaly detection, improving robustness.

2) *TiT-Diffusion Model*: is a diffusion-based generative model enhanced with Transformer and iTransformer architectures, designed to effectively capture the intricate temporal and inter-variable dependencies in MTS. As illustrated in Fig. 2(b), the denoising module constitutes the core component of the diffusion process. The residual blocks shown in Fig. 2(c) serve as the primary representation learning units for modeling both temporal dynamics and inter-variable dependencies. In addition, the diffusion embedding module depicted in Fig. 2(d) provides step-aware conditioning, enabling the denoising process to adapt to different diffusion stages.

3) *Ensemble Strategy*: integrates multiple TiT-Diffusion models with distinct masking patterns, trained on the MTS imputation task to exploit temporal and inter-variable dependencies. This reduces uncertainty and improves anomaly detection performance. To enhance diversity and collaboration, a modulating factor is propagated across models, guiding each model to focus on regions poorly handled by previous ones, thereby improving robustness.

4) *Anomaly Detector*: achieves precise and robust anomaly detection by computing anomaly scores derived from the reconstruction errors of masked observed values. Additionally, during the inference process of the final TiT-Diffusion model, we utilize the step-by-step denoised outputs as valuable indicators for predicting anomalies.

B. TiT-Diffusion (Implementation Details of EnsDiffAD)

The backbone of EnsDiffAD is a diffusion model [11], which recovers the underlying data structure by progressively denoising heavily corrupted observations. While diffusion models are effective at modeling complex data distributions, their denoising performance can be unstable when temporal dynamics and inter-variable correlations are entangled within a single modeling mechanism. To address this issue, TiT-Diffusion introduces a structured denoising backbone that explicitly separates the modeling of temporal and inter-variable dependencies. In this section, we provide the implementation details of the TiT-Diffusion.

1) *Denoising Diffusion Model*: Like other generative models, TiT-Diffusion aims to generate target data samples from noise. Let $\tilde{X}_t$ for $t = 1, \dots, T$ be the sequence of latent variables in the same sample space as $\tilde{X}_0$, which is denoted as $\tilde{X}$ in Section II-A1. Similar to traditional diffusion model, TiT-Diffusion consists of two modules: the noising module and the denoising module, corresponding to the forward and reverse processes respectively, both of which form a parameterized Markov chain.

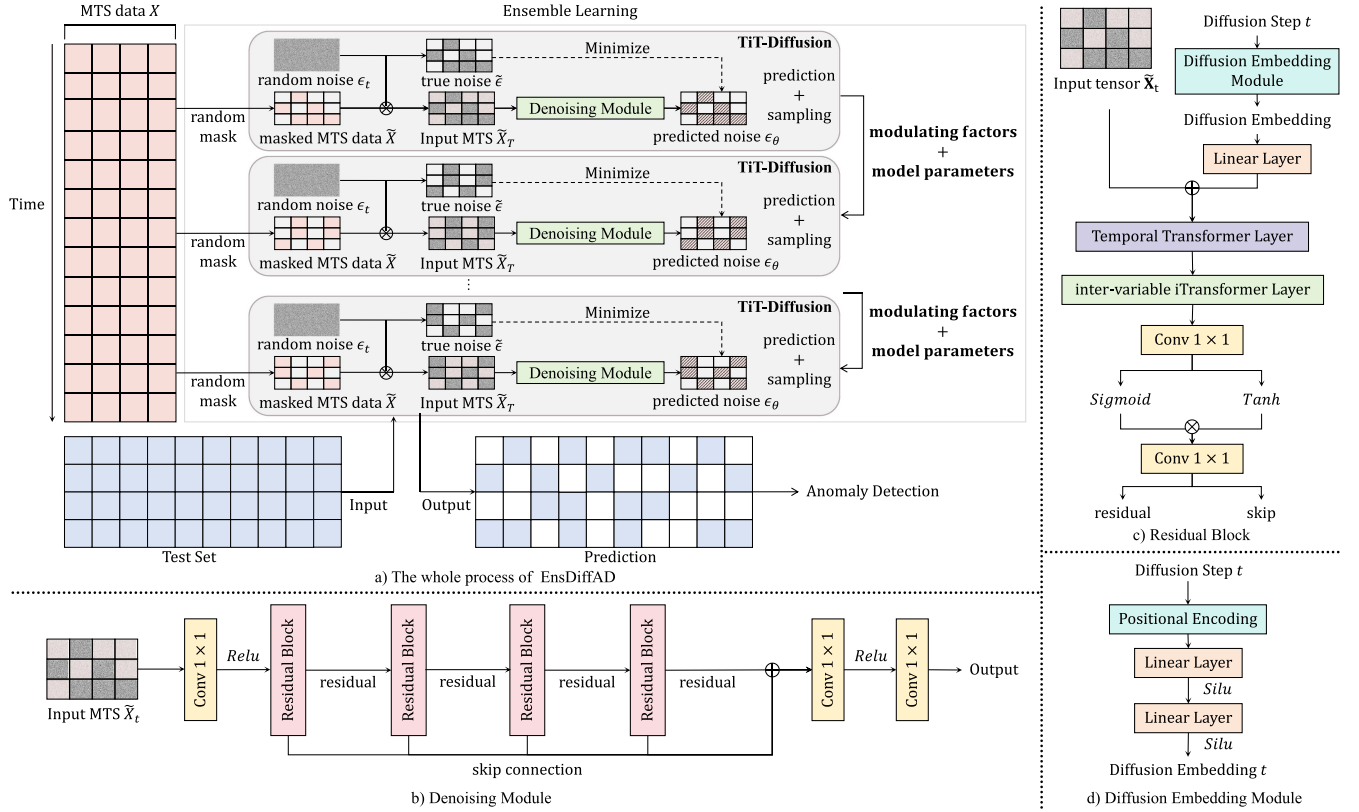


Fig. 2. Overview of EnsDiffAD Framework.

The nosing module, i.e., the process of adding noise, satisfies:

$$\begin{aligned} q(\tilde{X}_{1:T}|\tilde{X}_0) &= \prod_{t=1}^T q(\tilde{X}_t|\tilde{X}_{t-1}), \\ q(\tilde{X}_t|\tilde{X}_{t-1}) &= \mathcal{N}(\tilde{X}_t; \sqrt{1-\beta_t}\tilde{X}_{t-1}, \beta_t I). \end{aligned} \quad (2)$$

Here, $q(\tilde{X}_{1:T}|\tilde{X}_0)$ represents the distribution of noise samples after T steps. It should be noted that β_t refers to the predefined noise variance schedule, gradually increasing such that: $\beta_1 < \beta_2 < \dots < \beta_t < \dots < \beta_{T-1} < \beta_T$. The conditional distribution of $\tilde{X}_t$ given $\tilde{X}_0$ is:

$$q(\tilde{X}_t|\tilde{X}_0) = \mathcal{N}(\tilde{X}_t; \sqrt{\bar{\alpha}_t}\tilde{X}_0, (1-\bar{\alpha}_t)I), \quad (3)$$

where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$ and $\alpha_t = 1 - \beta_t$. Then, $\tilde{X}_t$ can be expressed as:

$$\tilde{X}_t = \sqrt{\bar{\alpha}_t}\tilde{X}_0 + \sqrt{(1-\bar{\alpha}_t)}\epsilon, \text{ where } \epsilon \sim \mathcal{N}(0, I). \quad (4)$$

Finally, $\tilde{X}_T$ will converge to standard Gaussian noise, serving as the input to the denoising module.

The denoising module, i.e., the reverse denoising process, involves recovering $\tilde{X}_{t-1}$ from $\tilde{X}_t$. The true posterior is Gaussian:

$$q(\tilde{X}_{t-1}|\tilde{X}_t, \tilde{X}_0) = \mathcal{N}(\tilde{\mu}_t(\tilde{X}_t, \tilde{X}_0), \tilde{\beta}_t I). \quad (5)$$

Since $\tilde{X}_0$ is unknown at inference, we approximate this distribution with a parameterized model:

$$\begin{aligned} p_\theta(\tilde{X}_{0:T}) &= p(\tilde{X}_T) \prod_{t=1}^T p_\theta(\tilde{X}_{t-1}|\tilde{X}_t), \\ p_\theta(\tilde{X}_{t-1}|\tilde{X}_t) &= \mathcal{N}(\tilde{X}_{t-1}; \mu_\theta(\tilde{X}_t, t), \Sigma_\theta(\tilde{X}_t, t)I). \end{aligned} \quad (6)$$

where $\tilde{X}_T \sim \mathcal{N}(0, I)$.

The Denoising Diffusion Probabilistic Model (DDPM) [11] estimates noise and predicts the mean value μ_θ by neural networks, thereby calculating $\tilde{X}_{t-1}$. The variance term is normally constant, which is not directly calculated through the neural network, but rather determined by the following diffusion process parameters:

$$\begin{aligned} \mu_\theta(\tilde{X}_t, t) &= \frac{1}{\sqrt{\alpha_t}} \left(\tilde{X}_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\tilde{X}_t, t) \right), \\ \Sigma_\theta(\tilde{X}_t, t) &= \sqrt{\tilde{\beta}_t}, \text{ where } \tilde{\beta}_t = \begin{cases} \frac{1-\alpha_{t-1}}{1-\alpha_t} \beta_t & t > 1 \\ \beta_1 & t = 1, \end{cases} \end{aligned} \quad (7)$$

where ϵ_θ is a trainable denoising function. DDPM utilizes ϵ_θ to estimate the noise vector that was added to its noisy input $\tilde{X}_t$. Under this parameterization, the reverse process can be trained by minimizing the difference between the true noise and the predicted noise. Since our model does not explicitly distinguish between masked and unmasked data during the denoising process, its ability to accurately impute missing values is limited. To address this limitation and enhance the model's ability to impute at masked positions, we propose the following optimization function:

$$\begin{aligned} \min_{\theta} \mathcal{L}(\theta) &= \min_{\theta} \mathbb{E}_{\tilde{x}_0 \sim q(\tilde{x}_0), \epsilon \sim \mathcal{N}(0, I), t} \\ &\frac{1}{|M_t|} \|(\epsilon - \epsilon_\theta(\tilde{x}_t, t)) \odot M_t\|_2^2, \end{aligned} \quad (8)$$

where $M_t = M_o - M$ indicates the target positions to be imputed by the model, M_o is the observation mask of the original MTS, and $\odot$ represents element-wise multiplication.

For the imputation, we apply a position-wise selection based on the input mask, retaining only the predicted values corresponding to the originally masked positions. The imputed result is thus obtained as $\hat{X}_{imputed} = \hat{X} \odot M_t$, where $\hat{X}$ is the sample generated by TiT-Diffusion. It is worth highlighting that diffusion models have shown significant competitiveness for imputation [36].

2) *Implementation Details of TiT-Diffusion*: In this section, we detail the implementation of TiT-Diffusion. The denoising module employs a neural network to predict the noise added at each diffusion step t and progressively reconstructs the data from $\tilde{X}_T$. As illustrated in Fig. 2 (b), this neural network is composed of multiple residual blocks and convolutional layers. First, the noise-added $\tilde{X}_t$ is projected into an internal feature space with C channels using a 1×1 convolution, forming the input tensor $\mathbf{X}_t \in \mathbb{R}^{C \times L \times K}$ for the residual blocks. Then, we take the first residual block as an example to illustrate the data processing procedure.

Besides $\mathbf{X}_t$, the residual block also takes the diffusion step embedding $\mathbf{e}_t$ (shown as Eq.(9)) at step t as an adaptive input through the diffusion embedding module, as shown in Fig. 2 (c). $\mathbf{e}_t$ serves as contextual information for the diffusion, allowing the model to incorporate awareness of the current denoising step. Specifically, the diffusion embedding module includes a sin/cos-based positional encoding block, two linear layers interleaved with the *SiLU* activation function, as illustrated in Fig. 2(d).

$$\begin{aligned} \mathbf{e}_t &= \text{SiLU}(W_2 \cdot \text{SiLU}(W_1 \cdot \mathbf{PE}(t))) \in \mathbb{R}^{T \times D}, \\ \mathbf{PE}(t) &= [\sin(\omega_1 t), \dots, \sin(\omega_d t), \cos(\omega_1 t), \dots, \cos(\omega_d t)] \end{aligned} \quad (9)$$

where D is the dimensionality of the diffusion embedding with $d = D/2$, and $\mathbf{PE}(t)$ denotes the sinusoidal positional encoding of diffusion step t , constructed using sine and cosine functions with different frequencies.

As shown in Fig. 2(c), in the residual block, the diffusion embedding $\mathbf{e}_t$ is projected via a linear layer to match the channel dimension C , and then combined with the input $\mathbf{X}_t$ to produce the joint embedding representation:

$$\mathbf{X}_t \leftarrow \mathbf{X}_t + \text{Linear}(\mathbf{e}_t). \quad (10)$$

Then, the joint embedding vector $\mathbf{X}_t$ undergoes sequential processing by *temporal Transformer* and an *inter-variable iTransformer* to simultaneously capture temporal and inter-variable dependencies. The data processing workflow is illustrated below.

$$\mathbf{H}_1 = \text{iTransformer}(\text{Transformer}(\mathbf{X}_t)) \quad (11)$$

Unlike traditional Transformers [9], [37], [38], which model temporal dependencies by applying self-attention along the time dimension and treating each time step as a token, the iTransformer adopts an inverted attention paradigm. Specifically, iTransformer treats each variable as an independent token and applies self-attention across the feature dimension, explicitly modeling inter-variable dependencies within multivariate time series. Although both architectures share similar self-attention

structures, their attention mechanisms operate on fundamentally different axes, resulting in distinct modeling scopes and representational characteristics. This distinction enables the Transformer to effectively preserve temporal dependencies, while allowing the iTransformer to robustly capture inter-variable dependencies that are critical for multivariate structure recovery under noisy and masked conditions.

Following a convolutional layer, $\mathbf{H}_1$ is divided into two branches that are processed using the *Sigmoid* and *Tanh* activation functions, respectively. After combining the outputs of the two branches via element-wise multiplication, the outputs of the residual block, referred to as *residual* and *skip*, are derived based on the following equation.

$$[\text{residual}, \text{skip}] =$$

$$\text{Conv}(\text{Sigmoid}(\text{Conv}(\mathbf{H}_1)) \odot \text{Tanh}(\text{Conv}(\mathbf{H}_1))) \quad (12)$$

It is notable that the residual blocks in the denoising module are connected in a residual manner. Furthermore, the *skips* from all residual blocks are aggregated and subjected to a projection operation, after which they are concatenated with the output *residual* from the final residual block. Finally, the predicted noise ϵ_θ is output through two successive 1×1 convolution layers. The effectiveness of integrating the temporal Transformer and inter-variable iTransformer is quantitatively validated by the ablation study in Section IV-C, where removing the iTransformer component consistently leads to performance degradation across multiple datasets.

C. Ensemble Strategy

Ensemble learning improves robustness and generalization by aggregating multiple complementary learners. In the context of MTSAD, ensemble methods are not only particularly effective for capturing complex, or noisy patterns, but also alleviate non-stationary in model performance across specific time periods or variable dimensions.

Motivated by these advantages, we introduce an ensemble learning strategy specifically tailored for diffusion models within EnsDiffAD. Consider an ensemble comprised of N TiT-Diffusion models, denoted as $\{D_\theta^{(1)}, \dots, D_\theta^{(N)}\}$. During training, each TiT-Diffusion $D_\theta^{(i)}$ receives a distinct version of the same MTS input, with masking applied randomly and independently. This variability encourages each model to specialize in imputing different regions of missing data, thereby promoting representation diversity and reducing model redundancy.

Additionally, unlike conventional ensemble methods that rely on simple averaging, our strategy employs an adaptive co-learning mechanism coupled with a progressive error correction process. Specifically, the n -th TiT-Diffusion model minimizes the following weighted denoising objective.

$$\begin{aligned} \min \mathcal{L}^{(n)}(\theta) &= \min \mathbb{E}_{\tilde{x}_0 \sim q(\tilde{x}_0), \epsilon \sim \mathcal{N}(0, I), t} \\ &\left(w^{(n)} \cdot \frac{1}{|M_t^{(n)}|} \left\| \left(\epsilon^{(n)} - \epsilon_\theta^{(n)}(\tilde{x}_t, t) \right) \odot M_t^{(n)} \right\|_2^2 \right), \end{aligned} \quad (13)$$

where $\epsilon^{(n)}$ is the real noise introduced by the noising module, $\epsilon_\theta^{(n)}$ denotes the estimated noise derived by the denoising module, and $M_t^{(n)}$ is the target imputation matrix for the n -th TiT-Diffusion model, respectively. Importantly, the modulating factor $w^{(n)} \in [0, 1]$ is a scalar per training sample, controlling how much the current model focuses on samples that were hard to reconstruct for preceding ensemble members. Inspired by [39], we compute $w^{(n)}$ from the average confidence of the preceding models on the same sample. Let

$$p^{(i)} = \exp \left(-\frac{1}{|M_t^{(i)}|} \left\| \left(\epsilon - \epsilon_\theta^{(i)}(\tilde{x}_t, t) \right) \odot M_t^{(i)} \right\|_2^2 \right), \quad (14)$$

where $p^{(i)} \in (0, 1]$ serve as a soft confidence score: larger reconstruction error yields smaller confidence. Then, $w^{(n)}$ of the n -th TiT-Diffusion model is formulated as:

$$w^{(n)} = \left(1 - \frac{1}{n} \sum_{i=1}^n p^{(i)} \right)^\gamma, \quad (15)$$

where γ is a sharpening coefficient that controls the model's sensitivity to hard samples with high reconstruction error. By assigning greater weights to hard samples where prior models performed poorly, each TiT-Diffusion model learns from the overall data distribution while focusing more on sample segments that were under-reconstructed by preceding models. The proposed ensemble strategy provides three main advantages: (i) It facilitates targeted learning refinement by directing subsequent models to concentrate on samples inadequately captured by preceding models. (ii) It enhances inter-model collaboration through modulating factors that function similarly to soft attention mechanisms, dynamically shifting the learning focus among ensemble members. (iii) It improves overall robustness by fostering representational diversity among models, which strengthens generalization and reduces the risk of overfitting to specific masking configurations.

D. Anomaly Detection With EnsDiffAD

Unlike traditional methods that directly aggregate the outputs of base models, we employ the final model in the ensemble, which integrates the knowledge learned from the preceding models. This model imputes missing values in the test MTS input. During the inference phase, two complementary sets of masks are applied for imputation, ensuring full coverage of the data. Anomaly scores are computed as the reconstruction errors between the imputed values and their corresponding ground truth observations.

Inspired by [9], we aggregate various imputation results derived from different denoising steps t to determine the final anomaly label for each time point l . Although the final step typically yields the smoothest reconstruction, it may suppress subtle abnormal patterns. In contrast, anomalies tend to manifest as persistent reconstruction inconsistencies across multiple diffusion steps, rather than as a single large error at the end of the denoising process, which motivates multi-step aggregation. For

each diffusion step t , we first derive a step-level anomaly indicator $Y_t = \{y_{t,1}, \dots, y_{t,l}, \dots, y_{t,L}\}$ based on the reconstruction error E_t , where the indicator function maps reconstruction errors to binary anomaly evidence under a step-adaptive threshold. The process is formulated as:

$$Y_t = \mathbb{1} \left(E_t \geq \frac{\Sigma E_T}{\Sigma E_t} \cdot \Theta_T \right), \quad (16)$$

where Θ_T denotes the upper anomaly ratio of reconstruction errors at final step T . This operation identifies time points that exhibit abnormal reconstruction behavior at the current denoising stage. To determine the final anomaly prediction at each time point l , a voting mechanism is applied:

$$y_l = \mathbb{1} \left(\sum_t^T y_{t,l} \geq \varsigma \right), \quad (17)$$

where ς is the voting threshold, respectively. A time point is designated as a final anomaly if it accumulates more than ς votes for anomalousness across all denoising steps. In our implementation, ς is not independently tuned, but is adaptively determined by the number of sampled denoising outputs used for voting. To enhance inference efficiency while maintaining accuracy, we perform sampling every three steps over the final 30 denoising iterations for the voting procedure. Leveraging the fact that diffusion models generate predictions progressively, intermediate outputs can be used as auxiliary signals to strengthen EnsDiffAD.

IV. EXPERIMENTS

In this section, we evaluate our model, EnsDiffAD, with extensive experiments to answer the following research questions:

- *RQ1*: How does EnsDiffAD perform against other baselines in the task of MTSAD?
- *RQ2*: How effective is each specific design in EnsDiffAD?
- *RQ3*: How does the anomaly detection performance of EnsDiffAD vary with different anomaly ratios?
- *RQ4*: What impacts does the parameters of the ensemble strategy in EnsDiffAD have on its anomaly detection performance?
- *RQ5*: How does the proposed modulating factor influence the training dynamics of EnsDiffAD?
- *RQ6*: What is the inference efficiency of EnsDiffAD, and does the ensemble strategy introduce additional computational overhead?

A. Experimental Settings

1) *Datasets*: We use the following five real-world datasets for anomaly detection experiments : MSL [40], PSM [41], SMAP [42], SWaT [43] and GCP [44]. Every dataset is partitioned into a training set and a testing set, with only normal data used for training, in accordance with typical unsupervised anomaly detection settings. The test set contains both normal and anomalous sequences, with the ground truth labels provided for performance evaluation.

2) *Baselines*: We compare EnsDiffAD with 15 representative baselines for MTSAD, which are grouped into five categories based on their primary modeling paradigms: (i) *Classical methods*: IForest [32].. (ii) *Temporal network-based methods*: MSCRED [45], LSTM-AD [46], OmniAnomaly [47], BeatGAN [48], MAD-GAN [48]. (iii) *Inter-variable modeling methods*: InterFusion [49], GDN [6], MTAD-GAT [7]. (iv) *Transformer-based methods*: TranAD [8], TSINR [50], DADA [51]. (v) *Diffusion-based methods*: ImDiffusion [9], D³R [52], MODEM [53].

3) *Evaluation Metrics*: Following previous studies [9], we employ Precision (P), Recall (R), and F1-score to evaluate the anomaly detection performance of both the baseline methods and EnsDiffAD. In addition, we introduce the Average Sequence Detection Delay (ADD) metric [54] to assess the timeliness and responsiveness of each detection method. The ADD metric is defined as follows:

$$\text{ADD} = \mathbb{E}[t_d - t^* \mid t_d \geq t^*], \quad (18)$$

where t_d represents the time when the anomaly is detected, and t^* is the time when the anomaly actually starts. Furthermore, we adopt the Range-based Area Under the Precision-Recall Curve (R-AUC-PR) metric proposed in [55], which is a threshold-independent evaluation metric designed for range-based anomaly detection in time series.

4) *Implementation Details*: The proposed EnsDiffAD is implemented using the PyTorch framework [56] and trained on an NVIDIA RTX 4090 GPU. The reported results are averaged over five independent runs to ensure robustness and reliability. Table II summarizes the detailed training configurations, diffusion settings, and model hyperparameters. Specifically, the forward diffusion process adopts a quadratic noise schedule, where $\beta_1 = 1.0e - 4$ and $\beta_T = 0.5$ over $T = 50$ diffusion steps, with the noise variance gradually increasing over time. For anomaly detection, the detection thresholds are dynamically adjusted for different datasets. The upper anomaly ratio Θ_T for the GCP dataset varies across its subsets, while a fixed upper anomaly ratio of 2% is used for the remaining datasets. For the baseline models, their hyperparameters and detection thresholds are set according to the configurations reported in their original papers.

B. Anomaly Detection Performance (RQ1)

We report the precision, recall, and F1-score of EnsDiffAD and all baseline models on five real-world MTS anomaly detection datasets, as summarized in Table III. Notably, EnsDiffAD overall demonstrates exceptional performance in terms of all evaluation metrics, namely precision (93.53%), recall (98.99%), and F1 score (96.14%). It achieves the highest average scores across five datasets, surpassing the performance of the other baseline methods. More specifically, EnsDiffAD consistently outperforms classical methods (e.g., IForest) and temporal network-based methods (e.g., LSTM-AD, MSCRED, OmniAnomaly, BeatGAN, and MAD-GAN) across all datasets. These results indicate that the proposed diffusion-based denoising mechanism, coupled with temporal and inter-variable

TABLE II
HYPERPARAMETER SETTINGS OF ENSDIFFAD

Hyperparameter	Value
Learning rate	1.0e-3
Batch size	16
Mask ratio	0.5
Number of residual blocks	4
Hidden dimension	128
Number of diffusion steps	50
Initial diffusion parameter β_1	1.0e-4
Time-varying diffusion parameter β_T	0.5
Number of ensemble models	3
Upper anomaly ratio Θ_T	0.02
Voting threshold ς	10

modeling, provides a more robust representation of multivariate time series than temporal network-based methods. Compared with inter-variable modeling baselines (e.g., GDN, MTAD-GAT, and InterFusion), EnsDiffAD achieves consistently higher recall and F1-scores on most datasets. While these methods explicitly incorporate cross-variable dependency modeling, their performance is more sensitive to noise and distributional shifts. In contrast, EnsDiffAD leverages diffusion-based generation to stabilize structure recovery during detection, leading to improved sensitivity to true anomalies. EnsDiffAD also demonstrates more stable performance than Transformer-based baselines (e.g., TranAD, TSINR, and DADA), particularly in terms of recall. This suggests that decoupling temporal and inter-variable modeling within the diffusion process is more effective than directly relying on global attention or latent representations for multivariate anomaly detection. Among diffusion-based baselines (ImDiffusion, D3R, and MODEM), EnsDiffAD still achieves superior overall performance. In particular, compared to the best-performing baseline (MODEM), EnsDiffAD improves precision, recall, and F1-score by 0.38%, 5.04%, and 3.54% respectively. This further validates the effectiveness of integrating the proposed TiT-Diffusion and ensemble strategy within a diffusion framework.

As shown in Table IV, EnsDiffAD also exhibits strong performance in terms of detection delay, measured by the ADD metric. It achieves the shortest delay on three datasets and secures the second-best performance on the remaining two. The incorporation of a random masking strategy to simulate missing data during training enables EnsDiffAD to adapt swiftly to diverse datasets, thereby effectively reducing computational latency. Compared to other methods, EnsDiffAD demonstrates a quicker response to anomalies and offers near real-time detection results. This capability renders it particularly suitable for online anomaly detection applications, where rapid response is essential.

C. Ablation Study (RQ2)

To better understand the design rationale of EnsDiffAD, we conduct ablation studies from three perspectives:

- *Architectural component analysis*: We first investigate the contribution of each core module by comparing: (1) the full model (EnsDiffAD), (2) a variant without the inter-variable iTransformer (w/o iTransformer), (3) a variant without the ensemble strategy (w/o ensemble), and (4) a variant

TABLE III

PERFORMANCE COMPARISON OF ANOMALY DETECTION METHODS ON FIVE REAL-WORLD TIME SERIES DATASETS. PRECISION (P), RECALL (R), AND F1 ARE AVERAGED OVER FIVE RUNS. ‘-’ INDICATES MISSING RESULTS DUE TO UNAVAILABLE CODE.

Dataset	MSL			PSM			SMAP			SWaT			GCP			Average		
Metric	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
IForest	0.6059	0.5328	0.5334	0.6630	0.4919	0.5641	0.2886	0.7671	0.4163	0.9764	0.6650	0.7907	0.8055	0.7385	0.7370	0.6679	0.6391	0.6083
MSCRED	0.5008	0.6088	0.4899	0.9555	0.6857	0.7965	0.4107	0.8604	0.2712	0.4823	0.4065	0.4407	0.9754	0.9735	0.9712	0.6649	0.7070	0.5939
InterFusion	0.7688	<u>0.9464</u>	0.8442	0.9533	0.9128	0.9326	0.8788	0.7704	0.8204	0.8683	0.8530	0.8600	0.9361	0.9720	0.9092	0.8811	0.8909	0.8733
LSTM-AD	0.7330	0.5745	0.6378	0.9050	0.7707	0.8313	0.7841	0.5630	0.6533	<u>0.9925</u>	0.6737	0.8026	0.9591	0.9575	0.9553	0.8747	0.7079	0.7761
GDN	0.8668	0.8072	0.8360	0.8750	0.8385	0.8564	<u>0.9689</u>	0.5401	0.6936	0.1311	0.0585	0.0808	0.9648	0.9628	0.9589	0.7613	0.6414	0.6851
OmniAnomaly	0.8321	0.8125	0.8221	0.9551	0.8859	0.9191	0.8407	0.9674	0.8995	0.9749	0.7500	0.8470	0.9572	0.9796	0.9668	0.9120	0.8791	0.8909
BeatGAN	0.7782	0.8512	0.8102	0.9204	0.8767	0.8975	0.8915	0.6781	0.7663	0.9606	0.7020	0.8107	0.9865	0.9630	0.9717	0.9074	0.8142	0.8513
MAD-GAN	0.7047	0.7841	0.7423	0.8596	0.8838	0.8698	0.9547	0.5474	0.6952	0.7918	0.5423	0.6385	0.9766	0.9558	0.9605	0.8575	0.7427	0.7813
MTAD-GAT	0.7321	0.7616	0.7432	0.8763	0.8725	0.8744	0.9718	0.5259	0.6824	0.8468	0.8224	0.8344	0.9490	0.9523	0.9461	0.8752	0.7869	0.8161
TranAD	<u>0.8951</u>	0.9297	<u>0.9115</u>	0.9506	0.8951	0.9220	0.8224	0.8502	0.8360	0.7025	0.7266	0.6886	0.9472	0.9812	0.9631	0.8636	0.8766	0.8642
ImDiffusion	0.8930	0.8638	0.8779	0.9811	0.9753	<u>0.9781</u>	0.8771	0.9618	0.9175	0.8988	0.8465	0.8709	0.9771	<u>0.9825</u>	<u>0.9774</u>	0.9254	0.9260	0.9244
TSINR	0.8357	0.8540	0.8447	0.9921	0.8937	0.9404	0.9167	0.7642	0.8335	0.9931	0.7232	0.8369	0.9216	0.8906	0.9053	<u>0.9318</u>	0.8251	0.8722
D ³ R	0.7848	0.9433	0.8567	0.6294	0.9619	0.7609	0.8680	0.9176	0.8921	0.7206	0.8529	0.7812	0.7515	0.9769	0.8666	0.7509	0.9305	0.8315
MODEM	0.9128	0.8832	0.8977	0.9697	<u>0.9835</u>	0.9776	0.8850	<u>0.9722</u>	<u>0.9266</u>	0.8942	0.9308	<u>0.9121</u>	-	-	-	0.9154	<u>0.9424</u>	<u>0.9285</u>
DADA	0.6870	0.9151	0.7848	0.7431	0.9211	0.8226	0.6585	0.8825	0.7542	0.6159	<u>0.9459</u>	0.7460	-	-	-	0.6761	0.9162	0.7769
EnsDiffAD	0.8813	0.9730	0.9249	<u>0.9715</u>	0.9977	0.9844	0.8799	0.9864	0.9301	0.9629	0.9948	0.9787	<u>0.9808</u>	0.9976	0.9890	0.9353	0.9899	0.9614

TABLE IV

COMPARISON OF AVERAGE DETECTION DELAY (ADD, MEAN± STD) OF ANOMALY DETECTION METHODS ON FIVE DATASETS

Dataset	MSL	PSM	SMAP	SWaT	GCP	Average
IForest	123±28	191±17	394±93	539±20	203±3	290±32
MSCRED	109±30	218±35	622±48	1065±339	145±3	432±91
InterFusion	32±15	40±10	423±4	454±141	141±1	218±34
LSTM-AD	115±29	224±54	541±51	627±4	107±1	323±28
GDN	106±2	148±0	402±4	1478±0	125±0	452±1
OmniAnomaly	93±2	121±11	116±38	550±48	131±6	202±21
BeatGAN	68±24	166±11	345±23	607±6	130±13	263±15
MAD-GAN	88±0	122±2	404±20	926±337	157±0	339±72
MTAD-GAT	96±17	182±0	542±2	482±80	145±3	289±20
TranAD	56±12	127±4	291±2	657±246	104±11	247±55
ImDiffusion	46±4	28±1	98±31	350±43	75±1	<u>120±16</u>
TSINR	310±9	156±5	<u>65±1</u>	391±23	200±4	224±8
D ³ R	90±12	172±6	514±32	643±46	105±0	305±19
MODEM	52±5	<u>26±0</u>	102±8	<u>343±26</u>	-	131±10
DADA	-	-	-	-	-	-
EnsDiffAD	<u>43±5</u>	21±3	47±11	52±6	<u>75±11</u>	48±7

without step aggregation during anomaly detection (w/o step aggregation).

- *Ensemble strategy analysis:* Building upon the above, we further examine the effectiveness of the proposed ensemble strategy by comparing it with two alternative ensemble schemes. This study provides a more fine-grained understanding of how different integration mechanisms influence performance and demonstrates the superiority of our design.
- *Masking strategy analysis:* Finally, we investigate the impact of different masking strategies. This analysis evaluates the robustness and adaptability of EnsDiffAD under varying masking designs.

As shown in Table V, EnsDiffAD achieves the best performance in most cases across all five datasets, consistently outperforming its ablated variants. Removing either the iTransformer, ensemble strategy, or the step aggregation strategy leads to clear performance degradation. This trend is further confirmed by the average results. Specifically, excluding the iTransformer reduces the F1 score from 0.9614 to 0.9417, while removing the ensemble strategy further decreases it to

0.9226. Additionally, excluding the step aggregation strategy results in a drop in performance, highlighting its importance in the model. Meanwhile, both variants result in increased ADD, indicating slower and less reliable detection.

As shown in Table VI, EnsDiffAD consistently achieves strong performance across all five datasets, and in most cases outperforms the alternative ensemble schemes. In contrast, both Output Average and Score Average show noticeable performance gaps on datasets such as MSL and PSM, indicating the limitations of simple averaging strategies. A clearer comparison can be observed from the average results. EnsDiffAD achieves the highest F1 score (0.9614), outperforming Output Average (0.9452) and Score Average (0.9469). Meanwhile, it maintains favorable ADD and R-AUC-PR values, indicating more reliable and timely anomaly detection. These results demonstrate that the proposed ensemble strategy more effectively captures complementary information and leads to more robust performance.

As shown in Table VII, different masking strategies exhibit varying performance across datasets, and no single strategy consistently dominates, indicating that masking effectiveness is inherently dataset-dependent. Despite these variations, Random Mask (default) maintains consistently strong and balanced performance across all datasets. This is further reflected in the average results, where it achieves the best performance in terms of P, R, F1, and R-AUC-PR, while maintaining competitive ADD. These results highlight the robustness of EnsDiffAD to different masking designs and its adaptability to diverse data characteristics, while also confirming random masking as a reliable default choice for stable performance.

D. Performance by Varying Anomaly Ratio (RQ3)

In this section, we examine the detection performance of EnsDiffAD by varying the anomaly ratio from 0.5% to 3.0%. As illustrated in Fig. 3, EnsDiffAD achieves consistently strong performance across all datasets, with most of them reaching peak F1 scores at an upper anomaly ratio of 2.0%. Notably, EnsDiffAD maintains high F1 scores with minimal fluctuation over all datasets, demonstrating the model’s robustness to

TABLE V
ABLATION STUDY FOR ARCHITECTURAL COMPONENT ANALYSIS

Method	MSL					PSM					SMAP				
	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR
EnsDiffAD	0.8813	0.9730	0.9249	36.7	0.5278	0.9715	0.9977	0.9844	18.7	0.5317	0.8799	0.9864	0.9301	30.7	0.4920
w/o itransformer	0.8667	0.9370	0.9004	43.6	0.5097	0.9607	0.9753	0.9679	22.0	0.5236	0.8797	0.9771	0.9259	43.2	0.4922
w/o ensemble	0.8697	0.9168	0.8926	47.3	0.5183	0.9853	0.9613	0.9732	25.1	0.5058	0.8754	0.9717	0.9210	252.0	0.4881
w/o step aggregation	0.8631	0.9234	0.8922	40.4	0.5033	0.9641	0.9484	0.9584	39.5	0.5121	0.8774	0.9838	0.9302	19.4	0.4914

Method	SWaT					GCP					Average				
	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR
EnsDiffAD	0.9629	0.9948	0.9787	43.1	0.5041	0.9808	0.9976	0.9890	91.4	0.5295	0.9353	0.9899	0.9614	44.1	0.5139
w/o itransformer	0.9143	0.9558	0.9346	175.2	0.4992	0.9755	0.9845	0.9795	89.8	0.5194	0.9194	0.9659	0.9417	74.8	0.5088
w/o ensemble	0.9352	0.9028	0.9187	295.2	0.4962	0.9805	0.8443	0.8773	148.8	0.5198	0.9441	0.9200	0.9226	153.7	0.5076
w/o step aggregation	0.9468	0.9751	0.9608	31.6	0.4970	0.9807	0.9896	0.9848	112.4	0.5226	0.9264	0.9641	0.9453	48.7	0.5053

TABLE VI
ABLATION STUDY FOR ENSEMBLE STRATEGY ANALYSIS

Method	MSL					PSM					SMAP				
	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR
EnsDiffAD	0.8813	0.9730	0.9249	36.7	0.5278	0.9715	0.9977	0.9844	18.7	0.5317	0.8799	0.9864	0.9301	30.7	0.4920
Output Average	0.8632	0.9390	0.8993	44.6	0.5191	0.9642	0.9622	0.9630	29.3	0.5057	0.8766	0.9825	0.9265	30.8	0.4918
Score Average	0.8661	0.9528	0.9072	39.0	0.5142	0.9732	0.9768	0.9749	22.5	0.5120	0.8784	0.9858	0.9290	23.8	0.4920

Method	SWaT					GCP					Average				
	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR
EnsDiffAD	0.9629	0.9948	0.9787	43.1	0.5041	0.9808	0.9976	0.9890	91.4	0.5295	0.9353	0.9899	0.9614	44.1	0.5139
Output Average	0.8993	0.9947	0.9445	32.9	0.5014	0.9881	0.9977	0.9926	94.4	0.5172	0.9183	0.9752	0.9452	46.4	0.5070
Score Average	0.8955	0.9941	0.9425	30.9	0.5037	0.9826	0.9989	0.9807	80.7	0.5393	0.9192	0.9817	0.9469	39.4	0.5122

TABLE VII
ABLATION STUDY FOR MASKING STRATEGY ANALYSIS

Method	MSL					PSM					SMAP				
	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR
Random Mask (default)	0.8813	0.9730	0.9249	36.7	0.5278	0.9715	0.9977	0.9844	18.7	0.5317	0.8799	0.9864	0.9301	30.7	0.4920
Blocking Mask	0.8425	0.9515	0.8937	48.6	0.5106	0.9615	0.8975	0.9284	56.6	0.5308	0.8794	0.9916	0.9321	10.9	0.4998
Grating Mask	0.8616	0.9239	0.8916	44.1	0.5077	0.9738	0.9793	0.9765	29.4	0.5049	0.8792	0.9920	0.9322	11.2	0.5019

Method	SWaT					GCP					Average				
	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR	P	R	F1	ADD	R-AUC-PR
Random Mask (default)	0.9629	0.9948	0.9787	43.1	0.5041	0.9808	0.9976	0.9890	91.4	0.5295	0.9353	0.9899	0.9614	44.1	0.5139
Blocking Mask	0.9305	0.9772	0.9533	77.2	0.4932	0.9732	0.9812	0.9766	77.8	0.5191	0.9174	0.9598	0.9368	54.2	0.5107
Grating Mask	0.9650	0.9912	0.9779	55.4	0.5012	0.9830	0.9825	0.9822	79.4	0.5238	0.9325	0.9738	0.9521	43.9	0.5079

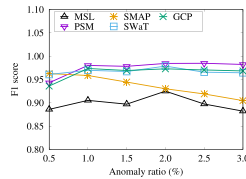


Fig. 3. F1 score to upper anomaly ratio variations on five datasets.

varying anomaly densities. On the other hand, when the upper anomaly ratio is too low (e.g., 0.5%), the model lacks sufficient anomaly signals during training, which tends to reduce recall and results in lower F1 scores. Conversely, excessively high upper anomaly ratios may introduce label noise or cause overexposure to ambiguous boundary regions, leading to increased false positives and decreased precision, and again lowering F1 scores

universally. This underscores that EnsDiffAD achieves optimal performance with moderate upper anomaly ratios, demonstrating robust stability and adaptability across datasets with diverse characteristics.

E. Impacts of Ensemble Parameters (RQ4)

To provide a comprehensive understanding of how the hyperparameters related to ensemble strategy affect the performance of EnsDiffAD, we conduct analyses from three perspectives:

- *Number of ensemble models*: We first evaluate the impact of the number of ensemble members on both detection performance and training efficiency (see Fig. 4 and Fig. 5).
- *Sharpening coefficient γ* : We further analyze the effect of the sharpening coefficient γ , which controls the strength of the modulating factor (see Fig. 6).

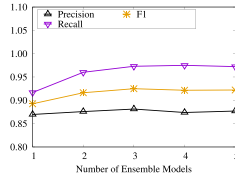


Fig. 4. Performance to number of ensemble models on MSL dataset.

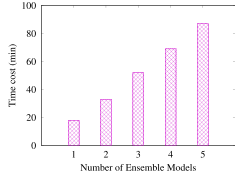


Fig. 5. Training time cost to number of ensemble models on MSL dataset.

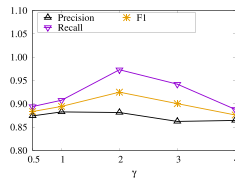
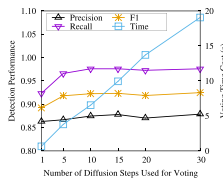
Fig. 6. Performance to γ on MSL dataset.

Fig. 7. Effect of Voting Step Number on MSL dataset.

- *Sampled denoising outputs for voting:* We finally analyze the influence of sampled denoising outputs used for voting aggregation (see Fig. 7). The voting threshold ς is determined by the number of sampled denoising outputs.

As shown in Fig. 4 and Fig. 5, model complexity and time cost increase with the number of ensemble models. It means more computational resources and longer training time are required. Detection performance initially improves, peaking at three models with the highest F1 score. Beyond that, performance plateaus or slightly declines, likely due to overfitting. Thus, a larger ensemble is not always beneficial; it may increase complexity, reduce training efficiency, and introduce a higher risk of overfitting. In this paper, we set the default number of ensemble models as three to achieve a good balance between anomaly detection accuracy and computational efficiency.

Fig. 6 shows the effect of the sharpening coefficient γ on the model's performance. As γ increases, recall improves, peaking at $\gamma = 2$, indicating enhanced sensitivity to true anomalies. However, further increases in γ lead to diminishing returns in recall, with a noticeable drop in performance at higher values.

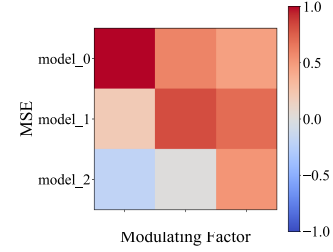


Fig. 8. Heatmap of reconstruction MSE and modulating weights on GCP subset at epoch 30.

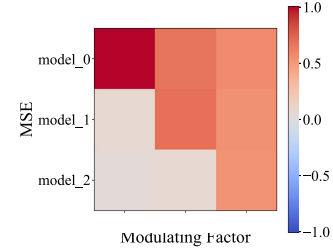


Fig. 9. Heatmap averaged over all training epochs on GCP subset.

The precision remains relatively stable and F1 score shows the same trend as recall. These trends align with the model's focus on harder-to-reconstruct samples, but excessive emphasis on these regions at higher γ values may lead to overfitting, disrupting the balance between recall and precision. Therefore, $\gamma = 2$ is chosen to strike the optimal balance in our method, EnsDiffAD.

To evaluate the effect of the voting threshold ς , we vary the number of sampled denoising outputs used for voting, including the 1, 5, 10, 15, 20, and 30 diffusion steps. As shown in Fig. 7, the detection performance improves when more denoising outputs are incorporated and becomes stable after 10 voting steps. In contrast, the voting time cost increases continuously with the number of sampled steps. These results indicate that using 10 diffusion steps provides a favorable balance between detection accuracy and inference efficiency. Therefore, EnsDiffAD adopts the 10 diffusion steps for voting by default.

F. Training Behavior Analysis (RQ5)

To analyze the training dynamics of the proposed ensemble strategy, we visualize the relationship between reconstruction errors and the modulating weights during training. Fig. 8 shows the heatmap of reconstruction MSE and the corresponding model weights at epoch 30. Samples with larger reconstruction errors tend to receive larger modulating weights, indicating that the weighting mechanism emphasizes samples that remain difficult for the current model. Fig. 9 presents the heatmap averaged over all epochs. The consistent patterns demonstrate that the modulating factor continuously assigns larger importance to samples with higher reconstruction errors throughout training. Fig. 10 illustrates the masked MSE curves of the three ensemble members. All models converge stably, and later ensemble members achieve lower reconstruction errors than earlier ones. This behavior suggests that the proposed reweighting mechanism

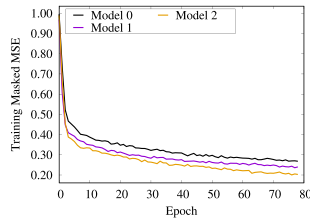


Fig. 10. Training MSE of the three ensemble members on GCP subset.

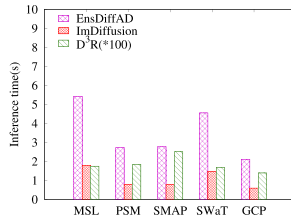


Fig. 11. Inference time comparison on five datasets.

encourages subsequent models to focus on difficult samples and progressively improve the reconstruction capability of the ensemble members.

G. Inference Efficiency Analysis (RQ6)

Although diffusion models typically require multi-step denoising during inference, the practical inference cost of EnsDiffAD remains manageable. The ensemble strategy is only applied during training, while inference uses only the final trained TiT-Diffusion model. Therefore, the inference complexity does not scale with the number of ensemble models. We further measure the average inference time per batch on an NVIDIA RTX 4090 GPU and compare it with representative diffusion-based baselines. The results (Fig. 11) show that EnsDiffAD maintains practical runtime while achieving superior detection performance. Although EnsDiffAD achieves favorable detection performance, its iterative denoising process still introduces more inference cost. Future work will explore model distillation, parallel sampling, reduced denoising steps, and mixed-precision inference to further reduce latency for real-time industrial deployment.

V. CONCLUSION

This paper introduces EnsDiffAD, a novel framework that employs several innovative strategies to enhance the accuracy and timeliness of multivariate time series anomaly detection significantly. EnsDiffAD integrates both temporal transformers and inter-variable transformers into a denoising network to optimize the diffusion model, thereby effectively capturing complex temporal and inter-variable dependencies in multivariate time series. Furthermore, we propose a novel loss function for the ensemble strategy that directs the model's attention to challenging regions through modulating factors. Extensive experiments on five publicly available multivariate time series anomaly detection

datasets demonstrate that EnsDiffAD significantly outperforms state-of-the-art anomaly detection methods.

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Qian Ma received the BS, MS, and PhD degrees in computer science and technology from Northeastern University, Shenyang, China, in 2011, 2013, and 2019, respectively. She is currently an associate professor with the College of Information Science and Technology, Dalian Maritime University, Dalian, China. Her research interests include data quality and Big Data management.



Shikai Guo received the BS degree in computer science from Liaoning University, in 2011 and the PhD degree in computer applications from Dalian Maritime University in 2018. He is currently an associate professor with Dalian Maritime University. His research interests include intelligent software engineering, EDA software optimization, and AI chip design.



Yanyang Li is currently working toward the undergraduation degree with the School of Information Science and Technology, Dalian Maritime University, Dalian, China, and pursue advanced studies. Her research interests include anomaly detection and related topics in data mining and machine learning.



Yu Gu received the PhD degree in computer software and theory from Northeastern University, China, in 2010. He is currently a professor with Northeastern University, China. His research interests include Big Data processing, spatial data management and graph data management. He is a senior member of China Computer Federation (CCF).



Mei Bai received the PhD degree in computer science and technology from Northeastern University, China, in 2016. She is currently an associate professor with Dalian Maritime University. Her research interests include data management, cloud computing, and query processing and optimization.



Xite Wang received the PhD degree in computer science and technology from Northeastern University, China, in 2015. He is currently an associate professor with Dalian Maritime University. His research interests include Big Data management and parallel data processing.



Ge Yu received (Senior Member, IEEE) the PhD degree in computer science from Kyushu University of Japan, in 1996. He is currently a professor with Northeastern University of China. His research interests include distributed and parallel database, OLAP and data warehousing, data integration, and graph data management. He is a fellow of CCF and member of the IEEE Computer Society, IEEE, and ACM.